

Cameron Zuschmidt

Philadelphia, PA | czusch26@engineering.upenn.edu | 717-696-8377 | linkedin.com/in/camzus

SUMMARY

Electrical engineering student (CS minor) with hands-on experience in hardware verification and debugging, embedded systems and avionics, PCB design, and power electronics. Seeking a Summer 2027 electrical engineering internship.

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science

Expected May 2029

B.S.E. in Electrical Engineering, Minor in Computer Science | GPA: 3.94/4.00

Philadelphia, PA

Relevant Coursework: Circuits and Systems, Computer System Architecture, Solid State Devices, Electricity and Magnetism, Programming Languages and Techniques

EXPERIENCE

Undergraduate Researcher – Hardware Verification

05/2026 – Present

Penn Instrumentation Group, High-Energy Physics

Philadelphia, PA

- Perform production quality-control testing on the Power and Timing Card (PTC), a Penn-designed readout board for the DUNE liquid-argon neutrino detector; validating all 250 production boards
- Debug I2C communication failures through MAX7357 mux trees to LTC2945 power/temperature monitors; support Zynq UltraScale+ SoC/FPGA bring-up under PetaLinux
- Develop Python test automation (WebSocket client/server architecture) covering PTC power, timing, sensor, and FPGA checks; contribute to the lab's automated board QC pipeline
- Configure a Beckhoff CX8110 EtherCAT master in TwinCAT and decode EtherCAT process data during board verification
- Authored technical documentation of the PTC architecture (Zynq UltraScale+ PS/PL partitioning, AXI interconnect, PetaLinux) and the DUNE readout chain; prepared a research poster on the QC effort

Executive Director and Operations Lead

06/2026 – Present

Penn Aerospace Club

Philadelphia, PA

- Lead a 150+ member club across jet propulsion, rocketry, high-altitude balloon, and aircraft design teams; manage a \$25,000+ budget, including fund allocation and procurement
- Organize speaker events, alumni panels, and industry mentorship connecting members with corporate recruiters

PROJECTS

Flight Computer and Avionics Bay, High-Power Rocketry Team (IREC)

09/2025 – Present

- Designed a custom STM32-based flight computer PCB in KiCad/Altium with IMUs, magnetometers barometer, radio, GPS, and flash logging; developed real-time sensor firmware under an RTOS
- Implementing a Multiplicative Extended Kalman Filter (MEKF) for quaternion attitude estimation, fusing gyroscope and accelerometer data via inverse-variance weighting
- Integrated a dual-deploy avionics bay with a consumer sensor suite of IMUs, altimeters, GPS, and ejection charges; reviewed the recovery architecture
- Driving control and actuation of airbrake system

FPGA-Based Digital Control of a Synchronous Buck Converter (In Progress)

08/2026 – Present

- Defined specs (12 V in, 5 V out, 3 A, 150 kHz) and simulated the synchronous buck power stage in LTspice
- Next: Verilog control loop on a Spartan-7 FPGA with PWM generation, dead-time insertion, synchronized ADC sampling, and digital compensator for closed-loop voltage regulation

Quantum Entanglement Experiment, MIT iQuHACK

01/2026

- Built a Qiskit experiment on IQM cloud quantum hardware demonstrating 20-qubit entanglement

LEADERSHIP & ACTIVITIES

Teaching Assistant, CIS 1100 (Python) – Committee Lead | **Treasurer**, ESE Student Advisory Board | **Avionics**, High-Powered Rocketry Subteam | **Avionics**, High-Altitude Balloon Subteam | **Board**, Penn Quantum Computing

SKILLS

Languages: Python, C, Verilog, Java

Hardware & EDA: Hardware testing and debugging, Altium Designer, KiCad, LTspice, schematic capture, PCB design

Embedded: STM32, Arduino, Zynq UltraScale+, PetaLinux, RTOS firmware, TwinCAT

Protocols & Tools: I2C, SPI, UART, EtherCAT, Ethernet, LoRa; Git/GitHub, Linux, Bash